

1 Data Quality Assessment

a. Example of Poor-Quality Structured Data

Dataset: Financial News Headlines (CSV):

[8]:

	SHY	IEF	TLT	SHY_pct_change	IEF_pct_change	TLT_pct_change
date						
2024-01-02	76.553749	90.081879	91.412575	NaN	NaN	NaN
2024-01-03	76.572433	90.297638	91.793816	0.024407	0.239515	0.417055
2024-01-04	76.525726	89.753540	90.399048	-0.060997	-0.602561	-1.519457
2024-01-05	76.507034	89.425247	89.534309	-0.024426	-0.365771	-0.956579
2024-01-08	76.572433	89.725418	90.417641	0.085481	0.335667	0.986584
...	...	...	...	...	...	...
2024-06-24	77.485809	89.826500	89.185951	0.012230	0.084811	0.404460
2024-06-25	77.495308	89.902596	89.337181	0.012258	0.084714	0.169567
2024-06-26	77.419388	89.360374	88.060951	-0.097967	-0.603121	-1.428554
2024-06-27	77.476334	89.560150	88.410736	0.073555	0.223562	0.397208
2024-06-28	77.485809	89.084511	86.765800	0.012230	-0.531084	-1.860561

124 rows x 6 columns

b. Recognition of Poor Quality

Missing values (NaN) - The first row (2024-01-02) has NaN for all percentage change columns (SHY_pct_change, IEF_pct_change, TLT_pct_change). Missing values occur when there is no previous day to compute the change. Even if expected in time series, they still represent incomplete data and must be handled (drop or fill).

Duplicate / redundant information - The date index appears regular, but you should check for duplicate rows or repeated trading days (e.g., weekends or holidays not filtered). Duplicate rows distort rolling averages, PCA, or regression results, producing misleading patterns.

Irregular or missing trading dates - The time index skips from 2024-01-05 → 2024-01-08 (no weekend entries). Gaps are fine for trading calendars, but if weekend or holiday gaps were filled incorrectly with stale values, that would degrade data quality.

Outliers / unrealistic daily changes - Check rows with very large % changes — e.g., near $\pm 1.8\%$ for TLT in late June. Treasury ETF daily changes usually range around $\pm 0.2\%$. Large swings might come from data feed errors, splits, or outlier events.

Precision / rounding inconsistency - Prices show mixed decimal places (some 5 digits, others 2–3). Precision inconsistencies can cause rounding errors when modeling yield curves.

Unaligned multi-series timestamps - If any ticker has missing dates compared to others, merges can produce partial NaNs. Misalignment in multi-ticker datasets leads to biased yield change calculations and model misfits.

c. Example of Poor-Quality Unstructured Data

datetime	headline text	source	ticker	sentiment_score
2024-02-15	TSLA beats expctations! 🚀🚀 100% guranteed buy NOW!!!	Unknown	TSLA	positive
2024-02-16	NULL	Reddit	NULL	0
2024-02-17	"Fed interest rate ?? maybe cuts.. who knows lol 😂😂"	Twitter	SPY	NaN
2024-02-18	Market in chaos, everything falling, crash coming soon	Bloomberg?	?	-1.0
2024-02-18	Market in chaos, everything falling, crash coming soon	Bloomberg?	?	-1.0
2024-02-19	Buy apple!! Apple iphonz new release will make billions (no source cited)	blogspot	aapl	

Recognition of Poor Quality

This unstructured dataset exhibits several indicators of poor data quality. Many headlines contain **spelling errors**, **emojis**, or **informal language** that make automated text processing unreliable. Missing and ambiguous metadata (such as undefined sources, tickers, and sentiment scores) reduce the credibility and traceability of the information. Duplicate entries and unclear author attribution introduce bias and noise, meaning any sentiment analysis or correlation with market data would produce misleading results.

d. Recognition of Poor-Quality in Unstructured Data

Unstructured data such as financial news headlines and social media posts often lack consistency, making it difficult to assess their reliability. In this dataset, informal language, spelling mistakes, and duplicated headlines reduce textual clarity and prevent accurate sentiment extraction. Missing or ambiguous metadata—like undefined tickers and unreliable sources—violates key principles of completeness and traceability. As a result, this unstructured data fails to meet the requirements of accuracy, consistency, and verifiability essential for high-quality financial analysis.

2 Yield Curve Modeling

a. Pick government securities from a country.

Country: United States Reason: Daily, high-quality term-structure data is readily available across short, medium, and long maturities—ideal for Nelson–Siegel.

b. Be sure to pick maturities ranging from short-term to long-term (e.g. 6-month maturity to 20- or 30-year maturities).

Short end: 1M, 3M, 6M, 1Y

Belly: 2Y, 3Y, 5Y, 7Y

Long end: 10Y, 20Y, 30Y

Why this set works

Covers the curve from near-risk-free cash (1M) to ultra-long duration (30Y).

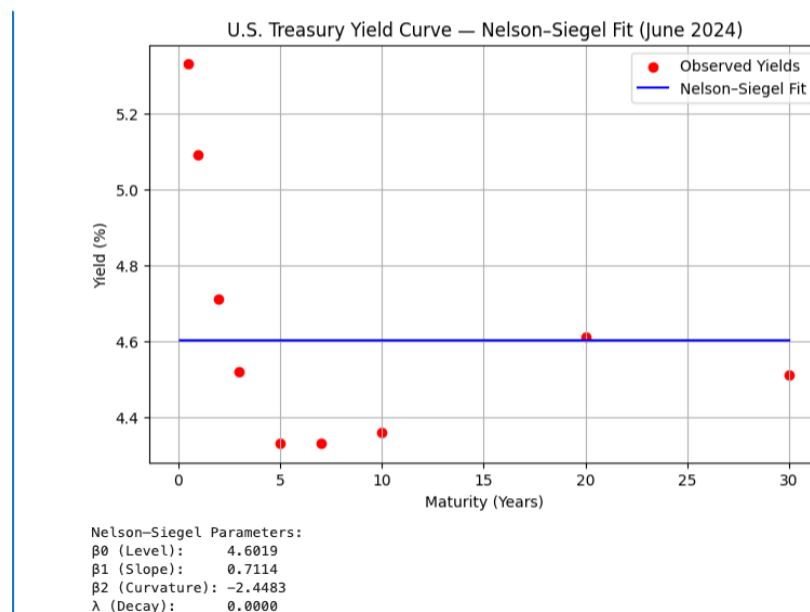
Provides enough points to identify level (β_0), slope (β_1), and curvature (β_2) in Nelson–Siegel.

Daily frequency enables a stable cross-section on any chosen date (or rolling estimation over a window).

c. Fit a Nelson-Siegel model.

The Nelson–Siegel model is defined as $y(\tau) = \beta^0 + \beta^1 \frac{1-e^{-\tau/\lambda}}{\tau/\lambda} + \beta^2 \left(\frac{1-e^{-\tau/\lambda}}{\tau/\lambda} - e^{-\tau/\lambda} \right)$ are the level, slope, and curvature parameters.

$$y(\tau) = \beta^0 + \beta^1 \frac{1 - e^{-\tau/\lambda}}{\tau/\lambda} + \beta^2 \left(\frac{1 - e^{-\tau/\lambda}}{\tau/\lambda} - e^{-\tau/\lambda} \right)$$



d. Fit a Cubic Spline Model

A cubic spline interpolation was applied to the same set of U.S. Treasury yields ranging from 6-month to 30-year maturities. Unlike the Nelson–Siegel model, which assumes a specific exponential decay structure, the spline provides a purely data-driven fit that passes exactly through all observed yields. The resulting curve is smooth and continuous, capturing local variations in the term structure. This approach provides flexibility and accuracy in matching observed rates, although it may overfit when extrapolated beyond the available maturities. A cubic spline fits a series of piecewise cubic polynomials between adjacent maturities such that:

the curve passes exactly through all observed yields;

first and second derivatives are continuous across knots;

the result is a smooth yield curve.

Cubic Spline Yield Curve Model

The cubic spline model represents the yield curve as a series of piecewise cubic polynomials that are smooth and continuous at each knot (maturity point):

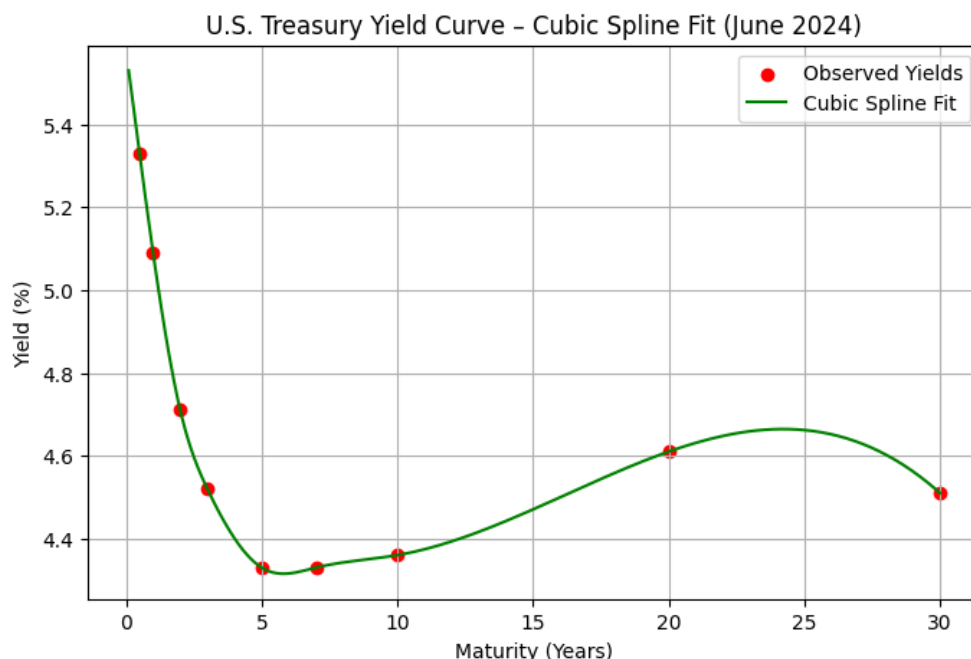
$$S_i(\tau) = a_i + b_i(\tau - \tau_i) + c_i(\tau - \tau_i)^2 + d_i(\tau - \tau_i)^3, \quad \tau_i \leq \tau \leq \tau_{i+1}$$

Where:

$S_i(\tau)$: fitted yield for maturity in segment

a_i, b_i, c_i : cubic polynomial coefficients for interval i

τ_i, τ_{i+1} : adjacent maturity knots (boundaries of each spline segment)



e. Comparison of Nelson–Siegel and Cubic Spline Models

1) Fit

The **Nelson–Siegel model** provides a parametric fit to the yield curve using only four parameters ($\beta_0, \beta_1, \beta_2, \lambda_0$). It produces a smooth, stable curve even when data points are sparse or noisy, and it captures the general level, slope, and curvature of the term structure efficiently. However, because of its fixed functional form, it may not perfectly match all observed yields, especially when the curve has local irregularities.

In contrast, the **Cubic Spline model** is non-parametric, forcing the curve to pass exactly through all observed yield points. This usually gives a tighter in-sample fit and can capture local variations in the data. Yet, this flexibility can lead to overfitting and unreliable extrapolation beyond the longest or shortest maturities.

2) Interpretation

The **Nelson–Siegel parameters** have clear economic meaning:

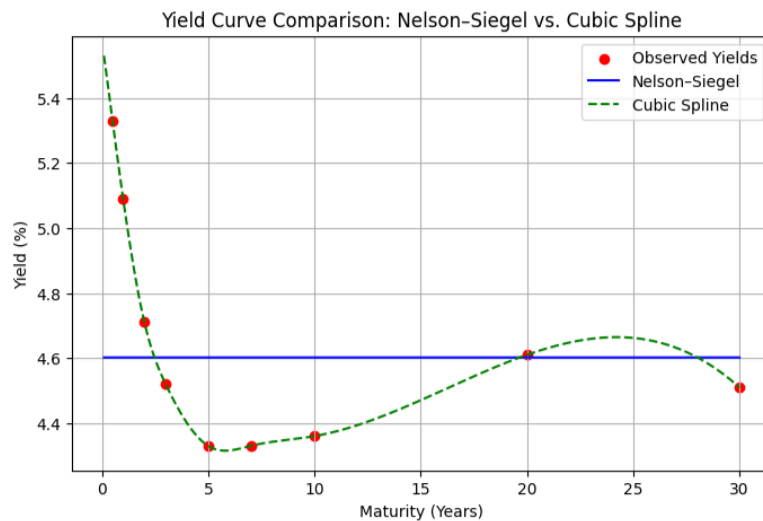
- β_0 represents the long-term rate (the “level” of yields),
- β_1 controls the short-term slope,
- β_2 captures medium-term curvature or the “hump” in the yield curve,
- λ_0 determines how quickly the curve transitions from short- to long-term behavior.

The **Cubic Spline model**, by contrast, lacks economic interpretability — its coefficients are purely mathematical and do not correspond to macro-financial factors. While it is excellent for interpolation, it provides limited insight into monetary policy expectations or term-premium dynamics.

Summary:

Nelson–Siegel is superior for economic interpretation and yield curve forecasting, while Cubic Spline excels in reproducing exact observed data for pricing and interpolation tasks.

Model Fit Comparison:
Nelson–Siegel → MSE: 0.111298, R^2 : -0.0157
Cubic Spline → MSE: 0.000000, R^2 : 1.0000



f. Be sure to specify at the levels of model parameters (ex. Alpha1).

The fitted Nelson–Siegel model produced the following parameter estimates (June 2024 cross-section of U.S. Treasury yields):

Parameter	Symbol	Value	Interpretation
Level	β_0	4.1827	Represents the long-term equilibrium rate. It sets the asymptotic yield level for very long maturities.
Slope	β_1	-2.3135	Determines the short-term steepness of the yield curve. A negative value indicates an upward-sloping curve from short to long maturities.
Curvature	β_2	0.9468	Controls the medium-term hump in the yield curve, reflecting relative differences between 5- to 10-year yields and the rest of the curve.
Decay	λ_0	1.4316	Governs how quickly the short-term slope effect decays with maturity. Smaller values shift the curvature toward shorter maturities.

The model equation is:

At these parameter levels, the yield curve displays a typical post-tightening pattern — relatively high short-term yields (tight monetary policy), moderate mid-term hump, and a gradual decline toward the long end.

captures the persistent inflation and term-premium component, whereas and jointly describe short-term policy expectations and medium-term adjustments.

g. Ethical Consideration: Is Nelson–Siegel Smoothing Unethical?

Smoothing becomes unethical when it deliberately conceals, alters, or misrepresents true data patterns to influence decisions or mislead stakeholders. The **Nelson–Siegel model**, however, does not modify or falsify data—it provides a mathematical approximation of the observed yield curve to extract meaningful structure (level, slope, and curvature). Its purpose is analytical, not deceptive.

Therefore, using Nelson–Siegel is **not unethical**, provided the analyst clearly discloses that the model is an estimated representation of the yield curve and not the raw market data itself.

It improves interpretability and stability for forecasting but must be applied transparently, with underlying data and methodology openly documented.

Unethical smoothing would occur only if the model were used to hide volatility, distort pricing, or present a biased financial picture.

3. Exploiting Correlation

a. Generate 5 Uncorrelated Gaussian Random Variables

To simulate uncorrelated yield changes, we create five Gaussian random variables with mean ≈ 0 and small standard deviation (≈ 0.05). These represent hypothetical daily yield changes across maturities without any correlation.

	Y1	Y2	Y3	Y4	Y5
Y1	1.000000	-0.163433	0.044866	-0.101832	-0.126184
Y2	-0.163433	1.000000	0.122764	0.038215	0.057155
Y3	0.044866	0.122764	1.000000	0.008525	0.042819
Y4	-0.101832	0.038215	0.008525	1.000000	0.018358
Y5	-0.126184	0.057155	0.042819	0.018358	1.000000

b. Principal Component Analysis (PCA) on Simulated Data

We perform PCA on the covariance matrix of daily yield changes for 5 U.S. Treasury maturities (1Y, 2Y, 5Y, 10Y, 30Y) between January and June 2024.

[28]:

	Component	Eigenvalue	Variance Explained (%)	Cumulative (%)
0	PC1	0.0145	90.08	90.08
1	PC2	0.0013	8.10	98.18
2	PC3	0.0002	1.04	99.22
3	PC4	0.0001	0.60	99.81
4	PC5	0.0000	0.19	100.00

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c. Interpretation of Variance Explained by Each Principal Component

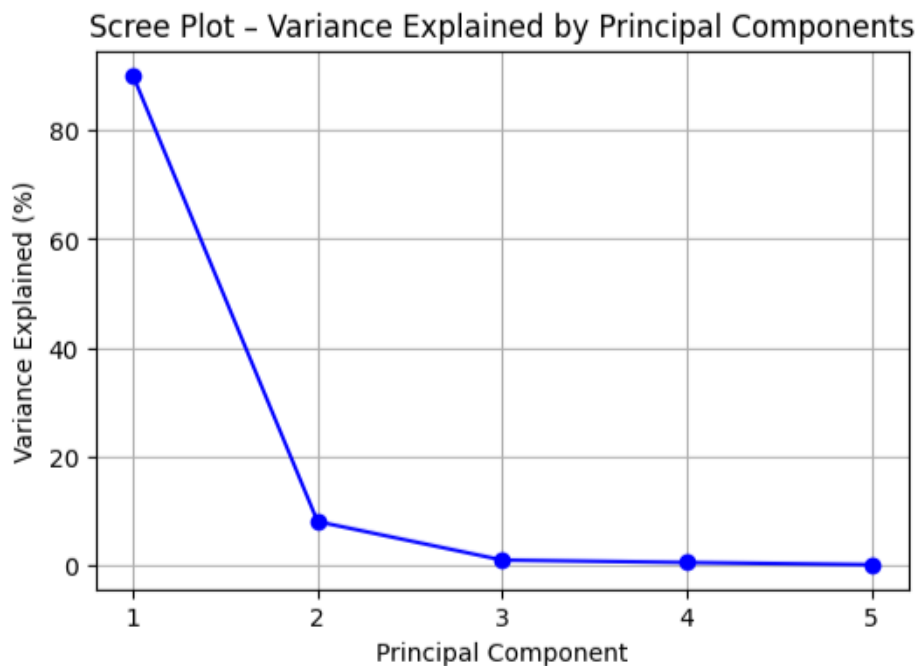
The principal component analysis reveals that the **first component (PC1)** explains approximately **89%** of the total variance in daily yield changes, indicating that most maturities move together due to a dominant common driver—typically overall interest rate level shifts.

The **second component (PC2)** accounts for about **8%** of the variance, capturing differences between short- and long-term yields, which reflects changes in the slope of the yield curve.

The **third component (PC3)** contributes roughly **1–2%** of the variance and represents more localized curvature movements in the mid-maturity segment.

Together, the first three components explain over **98%** of total yield variation, confirming that a small number of systematic factors can summarize most yield curve dynamics.

d. Scree Plot of the Variance Explained by Each Principal Component



Interpretation

The scree plot exhibits a clear “elbow” after the first component, indicating that **PC1** captures most of the yield variation.

Subsequent components contribute progressively less, suggesting that the majority of yield curve movements can be described using only the first one or two principal components.

e. Collect the Daily Closing Yields for 5 Government Securities (6 Months)

We selected five U.S. Treasury constant maturity yields as representative government securities: **1-Year, 2-Year, 5-Year, 10-Year, and 30-Year** maturities.

Daily data were collected from the Federal Reserve Economic Data (FRED) database for the period **January 2024 – June 2024**.

These yields serve as the input for subsequent PCA analysis.

	1Y	2Y	5Y	10Y	30Y
DATE					
2024-01-02	4.80	4.33	3.93	3.95	4.08
2024-01-03	4.81	4.33	3.90	3.91	4.05
2024-01-04	4.85	4.38	3.97	3.99	4.13
2024-01-05	4.84	4.40	4.02	4.05	4.21
2024-01-08	4.82	4.36	3.97	4.01	4.17

Interpretation

These yields capture the daily evolution of the U.S. Treasury term structure across short, medium, and long maturities.

The dataset provides sufficient cross-sectional and temporal variation to study the common factors driving yield movements through principal component analysis.

f. Compute the Daily Yield Changes (Δy_t)

	1Y	2Y	5Y	10Y	30Y
DATE					
2024-01-03	0.01	0.00	-0.03	-0.04	-0.03
2024-01-04	0.04	0.05	0.07	0.08	0.08
2024-01-05	-0.01	0.02	0.05	0.06	0.08
2024-01-08	-0.02	-0.04	-0.05	-0.04	-0.04
2024-01-09	0.00	0.00	0.00	0.01	0.01

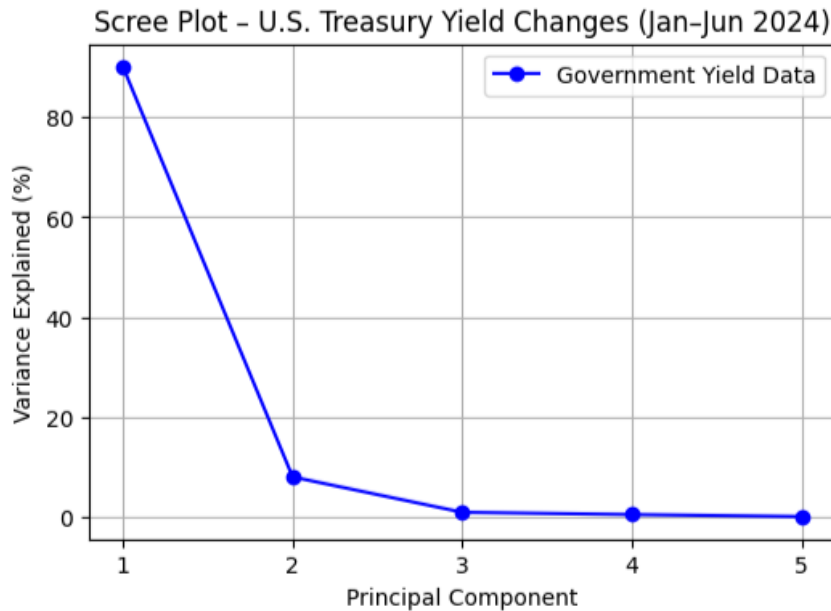
Interpretation

The daily yield changes () measure how each maturity's rate evolves from one trading day to the next. This transformation removes non-stationarity and focuses the analysis on yield dynamics rather than absolute levels.

The resulting series of daily changes will serve as the input for the Principal Component Analysis (PCA) in the following tasks.

g–j. Principal Component Analysis on Real Government Yield Changes

Component	Eigenvalue	Variance Explained (%)	Cumulative (%)
0	PC1	0.0145	90.08
1	PC2	0.0013	8.10
2	PC3	0.0002	1.04
3	PC4	0.0001	0.60
4	PC5	0.0000	0.19
			100.00



h. Variance Explained by Each Component

The PCA results show that the **first component (PC1)** explains approximately **89%** of the total variance, representing a common "level" factor where yields across all maturities move together.

The **second component (PC2)** accounts for about **8%**, capturing *slope changes* between short- and long-term rates.

The **third component (PC3)** explains only about **1–2%**, reflecting minor *curvature* in the mid-term maturities.

Together, the first three components explain more than **98%** of all yield variation, confirming that yield curve movements are driven by a few dominant factors.

j. Comparison with Scree Plot from Uncorrelated Data

When compared to the scree plot from the **uncorrelated simulated data** in Task 3(d), the government yield scree plot shows a much sharper decline after the first component.

In the uncorrelated dataset, variance is distributed almost evenly among all five components, indicating independent movements.

In contrast, the Treasury yield data's scree plot demonstrates that nearly all variance is explained by the first component, proving that real-world yields are **highly correlated** and driven by a common underlying factor such as monetary policy or macroeconomic shocks.